

IDone — Project Overview & Tech Stack

SYSTEM SPECIFICATION: Codebase Architecture, File Structure Inventory, Design System & API Reference

Project Name	IDone — Decentralized Identity Vault
Core Philosophy	«Your Identity. Your Credentials. Your Control.»
Framework Architecture	Next.js 14 App Router + FastAPI Asynchronous Microservice
Cryptographic Standard	W3C DID v1.0, W3C VC v1.1, RFC 8785 JCS, AES-256-GCM, Ed25519
Design Paradigm	Strict Zero-Gradients Policy Institutional Solid Color Tokens Accessible Micro-Animations
Automated Test Coverage	17 Passing Tests (100% Core Domain & Security Suite)
One-Click Launcher	launch.bat (Self-Healing Launcher) & stop.bat (Port Terminator)

1. Executive Summary & Core Philosophy

IDone is a privacy-first, enterprise-grade Decentralized Identity Vault engineered to combine the security of a modern hardware-backed digital wallet, the reliability and clarity of a top-tier banking interface, and the defensive rigor of a professional cybersecurity system.

The project strictly rejects the speculative, chaotic aesthetics of generic Web3 dashboards. Instead, it embodies a philosophy of institutional calm, mathematical precision, and uncompromising user sovereignty: identity data remains strictly off-chain, encrypted at the client boundary, and governed exclusively by the holder.

2. Comprehensive Technology Stack

Every layer of the IDone stack is selected for deterministic behavior, cross-platform reproducibility, and zero-compromise security.

Layer	Technology	Version / Standard	Architectural Role
Frontend UI	Next.js (App Router)	v14.1.0	React Server & Client Components, routing, SSR/SSG.
Language	TypeScript	v5.0+	Strict static type validation and client data integrity.
Styling	Tailwind CSS	v3.3+	High-trust solid palette with zero gradients.
Client Crypto	WebCrypto API	W3C crypto.subtle	Client-side AES-256-GCM and key derivation.
Icons	Lucide React	v0.344+	Clean, accessible institutional interface icons.
Backend API	FastAPI	v0.110+	High-performance asynchronous Python REST services.
API Runtime	Uvicorn	v0.28+	ASGI server powering asynchronous request pipeline.
ORM / Storage	SQLAlchemy	v2.0+	Connection-pooled PostgreSQL with resilient SQLite fallback.
Data Contracts	Pydantic	v2.6+	Strict input/output request serialization and schema validation.
Backend Crypto	cryptography (HAZMAT)	v42.0+	Ed25519 signing, RFC 8785 canonicalization, Argon2id.
Smart Contracts	Solidity	^0.8.20	IdentityRegistry.sol & CredentialStatus.sol on EVM.

Contract Deploy	Ethers.js / ts-node	v6.11+	Automated deployment and testnet interaction scripts.
Testing	pytest	v8.0+	17 automated backend, blockchain, and security tests.

3. The Solid Design System (Zero Gradients Policy)

IDone strictly prohibits all linear gradients, gradient meshes, and glowing overlays. In enterprise cybersecurity and Swiss banking, gradients are associated with speculative marketing and consumer tech hype. Solid colors project permanence, precision, and institutional stability.

Token Name	Hex Code	Tailwind Class	Semantic Purpose & Contrast Usage
Deep Navy	#0F172A	bg-navy-900 / text-navy-900	Primary text, headers, institutional hero backdrops.
Off-White	#F8FAFC	bg-offwhite-50	Canvas background, high-contrast structural separation.
Surface White	#FFFFFF	bg-white	Card surfaces, modal dialogs, data table backgrounds.
Trust Blue	#2563EB	bg-trust-600 / text-trust-600	Primary actions, DID indicators, active tab indicators.
Hover Blue	#1D4ED8	hover:bg-trust-700	Tactile button press and hover states.
Emerald Success	#16A34A	bg-success-600	Valid signatures, active credentials, optimal health score.
Crimson Alert	#DC2626	bg-alert-600	Revoked credentials, signature mismatches, critical alerts.
Amber Warning	#D97706	bg-amber-600	Pending key rotations, expiring credentials, audit notices.
Slate Gray	#64748B	text-slate-500	Secondary labels, timestamps, multihash identifiers.
Border Gray	#E2E8F0	border-offwhite-200	High-contrast structural dividers and card boundaries.

Calm Micro-Interactions & Animation Tokens:

- .btn-press: Instant subtle scale transform (active:scale-[0.98]) providing tactile button feedback.
- .card-interactive: 200ms translateY(-2px) elevation with crisp shadow on hover for cards.
- animate-calm-pulse: 2.4s calm opacity breathing for active cryptographic anchors.
- animate-fade-slide-up: 250ms cubic-bezier entry reveal for telemetry cards and audit records.
- @media (prefers-reduced-motion: reduce): Full accessibility compliance instantly disabling motion.

4. Complete Directory & File Structure Inventory

The repository follows a clean, modular structure without unnecessary subdirectories or redundant configuration files.

Relative File Path	Component Layer	Technical Responsibility & Key Exports
launch.bat	Operations	Self-healing one-click launcher for Windows (Python/Node check, auto .env, parallel boot, browser open).
stop.bat	Operations	One-click port terminator for PID lookup and process clearing on ports 8000 and 3000.
HOW_TO_PUSH_TO_GITHUB.txt	Documentation	Complete step-by-step Git initialization, remote setup, and GitHub push guide.
frontend/app/layout.tsx	Frontend Root	Global HTML structure, metadata, Inter font provider, and body wrappers.
frontend/app/globals.css	Design System	CSS variables, solid color tokens, .btn-press, .card-interactive, and custom scrollbar.
frontend/app/page.tsx	Public Landing	Hero section, live cryptographic verification preview card, trust badge, feature tiles.
frontend/app/dashboard/page.tsx	Dashboard	Command center, SecurityHealthCard, active DID badge, telemetry count tiles.

frontend/app/identity/page.tsx	Identity Hub	W3C JSON-LD DID document viewer, Ed25519 key rotation, assertion method updates.
frontend/app/credentials/page.tsx	Credentials Hub	VC management, issuance modal, external JSON-LD import, revocation controls.
frontend/app/vault/page.tsx	Encrypted Locker	Client-side AES-256-GCM locker, category filtering, in-browser decryption.
frontend/app/verify/page.tsx	Public Verifier	Independent verification tool with test vectors and multi-step checklist.
frontend/app/activity/page.tsx	Audit Trail	Security audit timeline, event filtering, cryptographic log history.
frontend/components/Navbar.tsx	UI Component	Top navigation bar with active network indicator dot and route links.
frontend/components/Sidebar.tsx	UI Component	Sidebar navigation with active indicator dots and hover transitions.
frontend/components/CredentialCard.tsx	UI Component	VC card with selective disclosure trigger, raw JSON modal, and revocation.
frontend/components/IdentityBadge.tsx	UI Component	DID badge with copy feedback tooltip and expandable W3C JSON-LD drawer.
frontend/components/SecurityHealthCard.tsx	UI Component	Animated solid score progress bar (700ms ease-out) and status checks.
frontend/components/ShareModal.tsx	UI Component	Selective disclosure dialog with individual attribute toggle switches.
frontend/components/VerificationResultCard.tsx	UI Component	Multi-step checklist with staggered reveals and report copy button.
frontend/lib/api.ts	Client API	Typed API client for all backend endpoints with JWT bearer injection.
frontend/lib/crypto.ts	Client Crypto	Browser WebCrypto utilities for AES-256-GCM encryption, decryption, and key derivation.
backend/app/main.py	Backend Entry	FastAPI initialization, CORS middleware, lifespan events, and route mounts.
backend/app/config.py	Configuration	Environment settings, JWT secrets, database connection URLs, and defaults.
backend/app/database.py	Database Engine	SQLAlchemy 2.0 session factory, connection pooling, and SQLite fallback.
backend/app/models.py	ORM Models	User, IdentityKey, Credential, VaultItem, and ActivityLog entities.
backend/app/schemas.py	Pydantic Schemas	Strict V2 data validation contracts for requests and responses.
backend/app/security.py	Crypto Engine	Argon2id hashing, AES-256-GCM helpers, Ed25519 signing, JWT encode/decode.
backend/app/services.py	Domain Services	DID generation, W3C document resolution, VC issuance, verification engine.
backend/app/routes/auth.py	API Route	User registration, login, and JWT bearer token issuance endpoints.
backend/app/routes/identity.py	API Route	DID document resolution (/identity/resolve) and key rotation endpoints.
backend/app/routes/credentials.py	API Route	VC issuance, listing, selective disclosure presentations, and revocation.
backend/app/routes/vault.py	API Route	Zero-knowledge encrypted locker CRUD operations (ciphertext storage).
backend/app/routes/verify.py	API Route	Dedicated public credential and presentation verification endpoint.
blockchain/IdentityRegistry.sol	Smart Contract	Solidity contract anchoring keccak256(did) hashes and controller addresses.
blockchain/CredentialStatus.sol	Smart Contract	Solidity contract maintaining on-chain revocation bitmaps and Merkle roots.
blockchain/deploy.ts	Deployment	Ethers.js v6 deployment script for Ethereum, Sepolia, Arbitrum, or Polygon.
tests/conftest.py	Test Harness	Pytest fixtures, isolated SQLite database, and test client configurations.
tests/backend/*.py	Unit Tests	Unit tests for auth, identity, credentials, vault, and verification.
tests/blockchain/test_contracts.py	Contract Tests	Tests for contract hash anchoring, revocation status, and event emission.
tests/security/test_security.py	Security Tests	Isolation tests, replay attack rejection, and zero plaintext leakage tests.
README.md	Master Guide	Full documentation with badges, architecture, setup, and GitHub push guide.

LICENSE	Legal	Standard MIT Open Source License.
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5. REST API Endpoint Specifications

The FastAPI backend provides a typed REST API documented via interactive Swagger UI at <http://127.0.0.1:8000/docs>.

HTTP Method	Path	Request Body / Params	Response Schema	Auth Level
POST	/auth/register	UserCreate (email, password)	UserResponse + AccessToken	Public
POST	/auth/login	UserLogin (email, password)	AccessToken (Bearer JWT)	Public
GET	/auth/me	None	UserResponse (id, email, did)	Bearer Token
GET	/identity/me	None	IdentityResponse (did, public_key_hex)	Bearer Token
GET	/identity/resolve/{did}	Path: did string	W3C JSON-LD DID Document	Public
POST	/identity/rotate-key	None	New IdentityKeyResponse	Bearer Token
GET	/credentials	Query: status, type	List[CredentialResponse]	Bearer Token
POST	/credentials/issue	CredentialIssueRequest	W3C Signed Verifiable Credential	Bearer Token
POST	/credentials/share	ShareRequest (credential_id, claims)	Verifiable Presentation Token	Bearer Token
POST	/credentials/{id}/revoke	RevokeRequest (reason)	RevocationConfirmation	Bearer Token
GET	/vault	Query: category	List[VaultItemResponse] (Ciphertext)	Bearer Token
POST	/vault	VaultItemCreate (ciphertext, iv, tag)	VaultItemResponse	Bearer Token
DELETE	/vault/{id}	Path: item id	DeleteConfirmation	Bearer Token
POST	/verify	VerifyRequest (credential or presentation)	VerificationResult (6 checks)	Public
GET	/activity	Query: limit, offset	List[ActivityLogResponse]	Bearer Token

6. Smart Contract Specifications

The smart contract architecture anchors cryptographic proofs on Ethereum-compatible chains without storing PII.

Contract	Interface Signature	State Impact & Event Emitted
IdentityRegistry	registerIdentity(bytes32 didHash, bytes32 docHash) external	Sets controllers[didHash] = msg.sender and documentHashes[didHash] = docHash. Emits IdentityRegistered.
IdentityRegistry	updateIdentity(bytes32 didHash, bytes32 newDocHash) external	Requires msg.sender == controllers[didHash]. Updates documentHashes. Emits IdentityUpdated.
IdentityRegistry	getIdentity(bytes32 didHash) external view returns (address, bytes32)	View query returning controller address and latest document root hash.
CredentialStatus	revokeCredential(bytes32 credHash) external	Requires msg.sender == owner. Sets isRevoked[credHash] = true and revocationTimestamps. Emits CredentialRevoked.
CredentialStatus	isCredentialActive(bytes32 credHash) external view returns (bool)	Gas-free read returning true if credHash is not revoked.

CredentialStatus	anchorRoot(bytes32 root) external	Updates rootAnchors accumulator. Emits RootAnchored for batch Merkle verification.
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7. Quality Assurance & Automated Test Suite (17 Tests)

The IDone test suite validates cryptographic soundness, API contract compliance, and security isolation across 17 automated tests:

Test Suite File	Test Case Function	Validation Focus & Invariant Verified
test_auth.py	test_register_and_login_flow	Verifies Argon2id password derivation, JWT token issuance, and expiration.
test_auth.py	test_register_duplicate_email	Guarantees 400 Bad Request on duplicate identity registration.
test_auth.py	test_register_password_mismatch	Ensures client registration password validation integrity.
test_identity.py	test_identity_lifecycle	Validates did:done: format, Ed25519 key generation, and DID document resolution.
test_credentials.py	test_credential_operations	Validates W3C VC issuance, Ed25519 signature, and selective disclosure claim filtration.
test_vault.py	test_vault_operations	Validates AES-256-GCM ciphertext creation, retrieval, and zero plaintext exposure.
test_verify.py	test_verify_valid_credential	Tests successful 6-step cryptographic verification of a valid credential.
test_verify.py	test_verify_tampered_credential	Ensures signature verification immediately fails if any claim field is modified.
test_verify.py	test_verify_expired_credential	Validates rejection of credentials beyond their expiration timestamp.
test_verify.py	test_verify_malformed_credential	Validates graceful rejection and error handling on corrupted JSON-LD schemas.
test_contracts.py	test_blockchain_hashing_and_anchoring	Tests keccak256 hash generation and IdentityRegistry state transitions.
test_contracts.py	test_credential_status_anchor_integrity	Tests revocation bitmap updating and CredentialStatus verification queries.
test_security.py	test_unauthorized_access_rejection	Validates HTTP 401 Unauthorized when accessing endpoints without bearer token.
test_security.py	test_forged_bearer_token	Confirms rejection of forged, tampered, or expired JWT tokens.
test_security.py	test_cross_user_isolation	Proves strict tenant isolation: User A cannot read User B's vault items or credentials.
test_security.py	test_no_private_keys_exposed	Guarantees private keys are never returned in API responses or logs.
test_security.py	test_argon2id_hash_format	Validates that password hashes in the database adhere strictly to the Argon2id standard.